

F657 — Verify the GSO piece (my flagged responsibility), then open the neutrino scale. Parity bank (K837) HOLDS — I checked my piece, including a Kramers concern, and it resolves. The one thing that could have broken “index=1 for both implementations” is: if $\mathbb{Z}$ is *antiunitary* (CPT/charge-conjugation), $\mathbb{Z}^2=-1$ triggers Kramers degeneracy — I worried that forces an *even* mode count, contradicting 1. It doesn’t: the Kramers pair $\{\psi_+, \psi_-\}$ is exactly {a Weyl mode, its CPT antiparticle} (ψ_- = the $(\bar{3}, 2, -1/6)_R$ conjugate of ψ_+) — NOT two independent physical states. So the Kramers “2-fold degeneracy” IS the particle–antiparticle pairing = one physical chiral Weyl. Unitary $\mathbb{Z}$ (project onto a $\pm i$ sector) also keeps one. Both implementations $\rightarrow$ one chiral fermion $\rightarrow$ index 1 $\rightarrow$ chiral. Elie’s “both give 1” is right; I concur, having actually run the Kramers check on the antiunitary branch I’d flagged. Parity banks. Then: neutrino scale (pull 23f) — Majorana is already derived ($Y=0$ real rep); the absolute scale is the open number; discrete recast ($\text{meV} / \exp(-280)$) is the right frame, with a hard target-innocence flag.

Lyra, Thu 2026-07-23. Signing off my flagged GSO piece (parity holds); opening the neutrino scale. After two sign errors I checked this by computing the Kramers branch, not estimating.

Parity bank — the GSO piece checked (I concur)

The bank rests on: $\mathbb{Z}^2=-1$ (Pin^- , ratified F656), index=1 (Elie), not-vector-like (Elie’s conjugate-rep). The piece flagged as mine — is $\mathbb{Z}$ unitary or antiunitary, and does the index stay 1 either way? - Unitary $\mathbb{Z}$ ($\mathbb{Z}^2=-1$): eigenvalues $\pm i$; the consistent projection $\frac{1}{2}(1 \mp i\mathbb{Z})$ onto one $\pm i$ sector keeps 1 of the 2-dim $\{\psi_+, \psi_-\}$. Index 1. $\mathbb{Z}$ - Antiunitary ($\mathbb{Z}$ (CPT, $\mathbb{Z}^2=-1$) — the Kramers worry: Kramers theorem forces 2-fold degeneracy for $T^2=-1$, which I worried gives an *even* count (0 or 2), not 1. Resolved: the degenerate pair $\{\psi_+, \psi_-\}$ is {Weyl mode, its CPT antiparticle} — a Weyl fermion and its antiparticle are ONE physical field, not two independent states. So the Kramers pair = the particle–antiparticle pair = one physical chiral Weyl. The count of *physical chiral fermions* is 1. $\mathbb{Z}$ Both branches $\rightarrow$ one chiral Weyl $\rightarrow$ index 1 $\rightarrow$ chiral. The only outcome that gave 0 was the invalid Pin^+ projection (my retracted F655). I concur with the bank — checked the antiunitary/Kramers branch I flagged, and it holds. Parity is derived: one chiral SM generation from {non-orientable boundary + $k=\pm 1$ instanton + $Y \neq 0$ complexity}.

The full derived structure (one column)

Reality-type read off (color, isospin, hypercharge): $Y \neq 0$ or colored $\rightarrow$ complex $\rightarrow$ chiral (parity); colored $\rightarrow \lambda_2 > 0 \rightarrow$ confined; $Y=0 \rightarrow$ real $\rightarrow$ Majorana. One structure: DIRAC bulk, squeeze, non-orientable boundary, isospin=weak-SU(2), $k=\pm 1$ instanton, SM charges, parity, confinement, custodial, ν -Majorana. The world is left-handed because its fermions carry hypercharge. Banked (given reps).

Neutrino absolute scale (pull 23f) — my contribution: structure + discipline

Majorana is DERIVED (F655: ν = lone $Y=0 \rightarrow$ real rep $\rightarrow$ modes pair $\rightarrow$ Majorana; Elie 4796). $m_1=0$ exact (domain rank-2, 2 idempotents $\rightarrow n(\nu_R)=2$). So the spectrum is $\{0, m_2, m_3\}$ and the sole open target is the *scale*. The discrete recast (Casey’s frame, and I endorse it): don’t grind the continuous seesaw $m_\nu \sim v^2/\Lambda$ — ask whether the neutrino scale is a discrete $\exp(-N)$ exponent like BST’s other scales, not a continuous cutoff. The natural handle: the meV coincidence. $m_\nu \sim$ tens of meV; $\Lambda_{DE}^{(1/4)} \sim$ few meV; and BST already gets $\Lambda = \exp(-280)$, $280 = 2^{\{N_c\} \cdot n_C \cdot g}$ (from the commitment-density structure). Both being meV is the known

ν -dark-energy coincidence — the recast asks if they share the $\exp(-280)$ discrete origin. ★ Target-innocence flag (hard, mine to hold): the neutrino mass is *measured* ($\Delta m_{32}^2 \sim 2.5 \times 10^{-3} \text{ eV}^2 \rightarrow m_3 \sim 50 \text{ meV}$). So any “ $m_\nu = (\text{BST integer}) \cdot \Lambda^{1/4}$ ” or “ $\exp(-N)$ with N a BST composite” is a fit-to-data unless the exponent N is derived *target-innocently* from the geometry (like 280 was). The failure mode: reading a BST-looking integer off the observed ratio $m_\nu/\Lambda_{\text{DE}}^{1/4}$ (~ 20 — and $20 = \text{rank}^2 \cdot n_C$ is exactly a multi-form trap from the flavor arc). Do NOT bank a discrete neutrino scale unless the exponent is derived before looking at the number. If no handle yields a target-innocent exponent, the honest answer is: the absolute scale is a continuous input (the seesaw Λ), and we say so plainly — a real boundary, not dressed up.

Tiers / handoffs

- @Elie — GSO piece checked and concurred: both unitary and antiunitary (Kramers) implementations give one chiral Weyl (the Kramers pair = particle+antiparticle, not two states). Parity bank holds. On the neutrino scale: the seesaw Λ / Weinberg-coefficient computation is yours — but recast discrete-first (does the scale drop out of the Bergman/heat-kernel $\exp(-N)$ structure?). Target-innocence: derive the exponent N from geometry, don’t read it off m_ν . If it’s continuous, report that plainly.
- @Keeper — I signed off the GSO/unitary-vs-antiunitary piece flagged as mine: the Kramers worry (antiunitary $\mathbb{Z}^2 = -1 \rightarrow$ even count) resolves because the degenerate pair is particle+antiparticle = one physical Weyl. Both implementations $\rightarrow$ index 1 $\rightarrow$ chiral. Parity bank K837 holds; I concur, having computed the branch, not rubber-stamped it. Neutrino: Majorana derived, $m_1=0$ exact, scale is the open number; discrete recast ($\text{meV}/\exp(-280)$) with the target-innocence flag as the gate.
- @Grace — parity holds (GSO checked); the unification figure’s parity cell = derived (Pin^- index 1, both implementations, Kramers-safe). Neutrino row: Majorana derived (your figure’s $Y=0$ real-rep cell); the scale is the open number, discrete-recast, target-innocence-gated. The $\text{meV}/\exp(-280)$ handle is a *lead* to render as candidate, not banked.
- @Casey — parity is genuinely banked, and I signed off the one piece that was mine to check rather than take it on faith: there was a real subtlety — if the boundary symmetry carries a charge-conjugation, a Kramers degeneracy could have forced an even count and spoiled the “one chiral fermion.” I worked it, and it resolves cleanly: the degenerate pair is just a Weyl mode and its own antiparticle — one physical fermion, not two — so the count is 1 either way. So the world-is-left-handed-because-it-carries-hypercharge result stands, on a checked sign and a checked count, not a lean. On the neutrino scale: Majorana falls out already, the spectrum is $\{0, m_2, m_3\}$, and the only open thing is the absolute scale. Your discrete-first instinct is right — the meV coincidence with dark energy (both $\sim \text{meV}$, and BST’s $\Lambda = \exp(-280)$) is the natural handle. But that’s exactly where I’ll be strictest on target-innocence: the neutrino mass is measured, so a discrete exponent only counts if we derive it from D_{IV}^5 *before* matching the number — otherwise it’s the flavor-arc fit-trap again (the ratio is $\sim 20 = \text{rank}^2 \cdot n_C$, which is precisely a multi-form coincidence). If a clean exponent yields, the scale is geometric; if not, I’ll tell you plainly it’s an honest continuous input.

Notes only; no toys/theorems claimed. GSO piece checked (Kramers-safe, both implementations $\rightarrow 1$) $\rightarrow$ parity bank HOLDS, I concur. Neutrino: Majorana derived, scale open, discrete recast ($\text{meV}/\exp(-280)$) gated hard on target-innocence. — Lyra